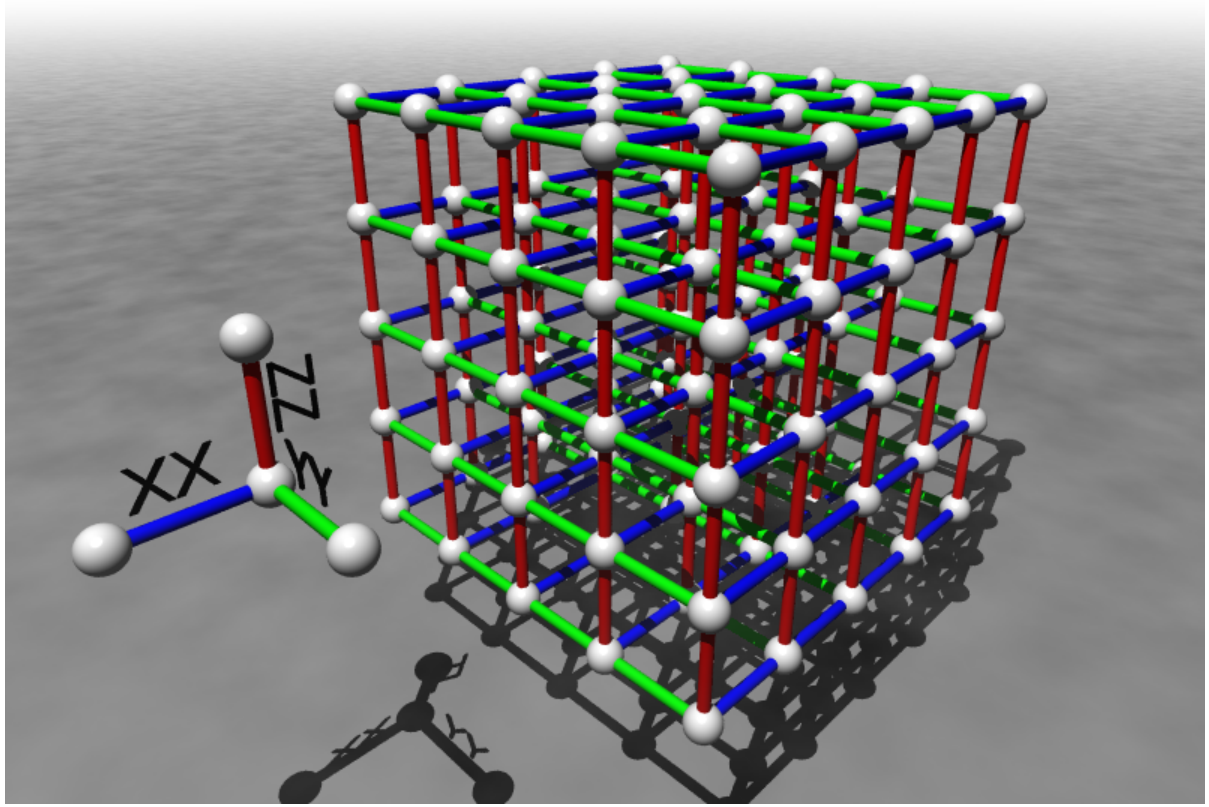
Novel Methods for Robust Quantum Computation



Dave Bacon

Department of Computer Science & Engineering

University of Washington

funded by NSF

1994

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

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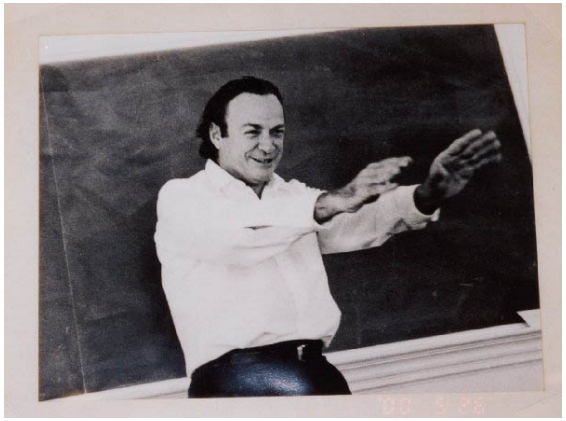
×

4727721461074353025362230719730482246329146953020971164598521711305207112563635903975 27



Peter Shor discovers that a computer which uses quantum data and quantum gates, can be used to efficiently factor and hence break the most widely used public key cryptosystems

Quantum Simulation



Richard Feynman

Simulated Quantum Computation of Molecular Energies

Alán Aspuru-Guzik,[1*†] Anthony D. Dutoi,[1*] Peter J. Love,[2] Martin Head-Gordon[1,3]

Science 9 September 2005:
Vol. 309. no. 5741, pp. 1704 - 1707

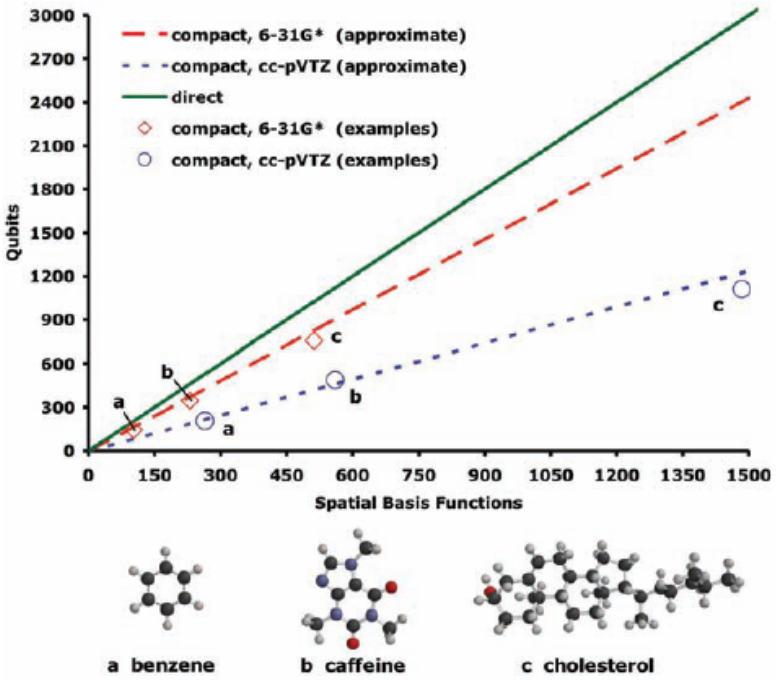
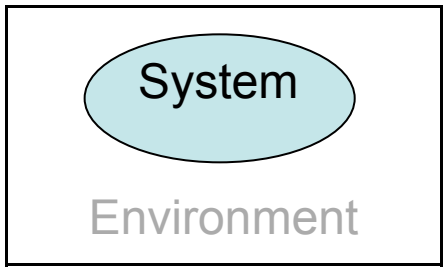


Fig. 1. Qubit requirements versus basis size. The number of qubits required to store the wave function of a molecule is shown as a function of the number of basis functions for different mappings. For the compact mapping, the qubit requirement depends also on the ratio of number of electrons to basis functions, which is relatively constant for a given basis set; although the higher quality cc-pVTZ basis is more economical per basis function, a molecule in this basis uses substantially more functions than with the 6-31G* basis. The qubits required for specific molecules and basis sets are also shown.

"For most molecules and basis set combinations shown in Fig. 1, an FCI calculation is certainly classically intractable…This demonstrates an often-stated conjecture, that quantum simulation algorithms with 30 to 100 qubits will be among the smallest applications of quantum computing that can exceed the limitations of classical computing."

The First Problems

THE DECOHERENCE PROBLEM



$$H = H_S \otimes I + I \otimes H_B + H_I$$

open quantum systems can become entangled with their environment

EXAMPLE:

evolve

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

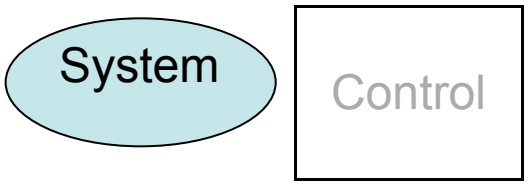
coherent superposition — evolve — incoherent sum

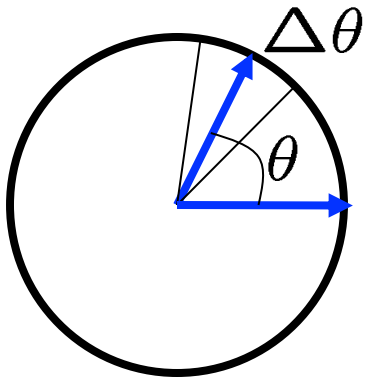
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

1. Decoherence turns quantum information into classical information
2. It does this on a time scale which only a few hundred to million times slower than the rate we can manipulate the quantum information

Second Problem

THE FAULTY CONTROL PROBLEM

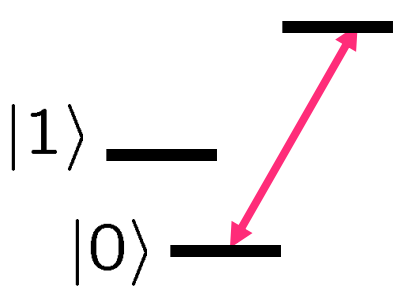




unitary gates cannot be implemented with arbitrary accuracy

THE FAULTY MEASUREMENT PROBLEM

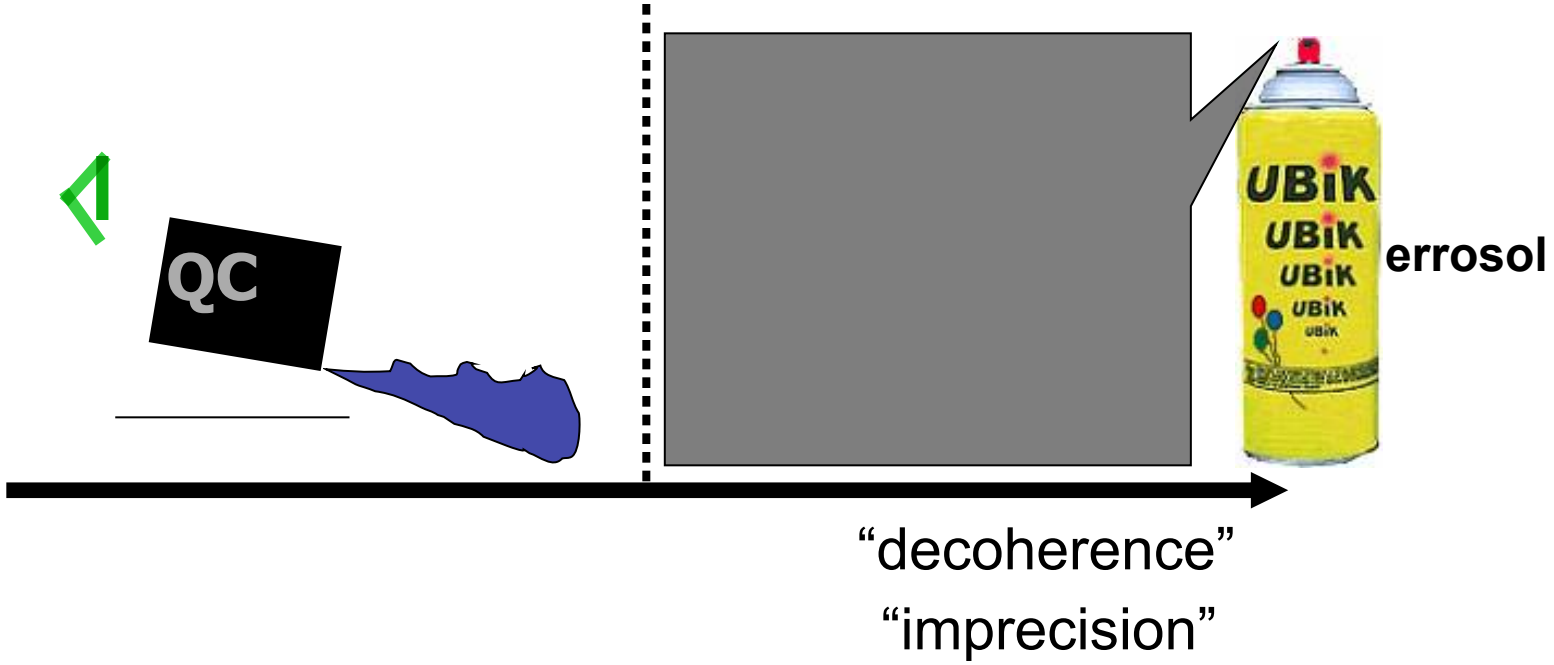
measurements are imperfect



wrong result, wrong resulting state

related state preparation problem

Problem Solution



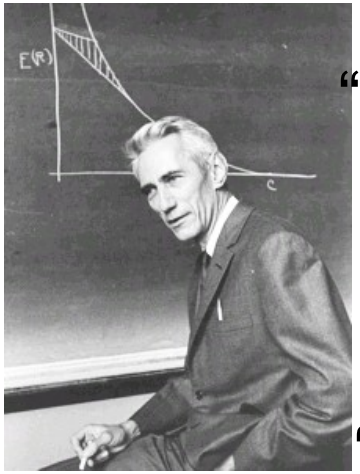
Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Kitaev, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

To Build a ~~Quantum~~ Computer
Classical

Problem (1938): How to build a reliable system from unreliable components?



Claude Shannon

"A mathematical theory of communication" (1948)

How to communicate in the presence of noise

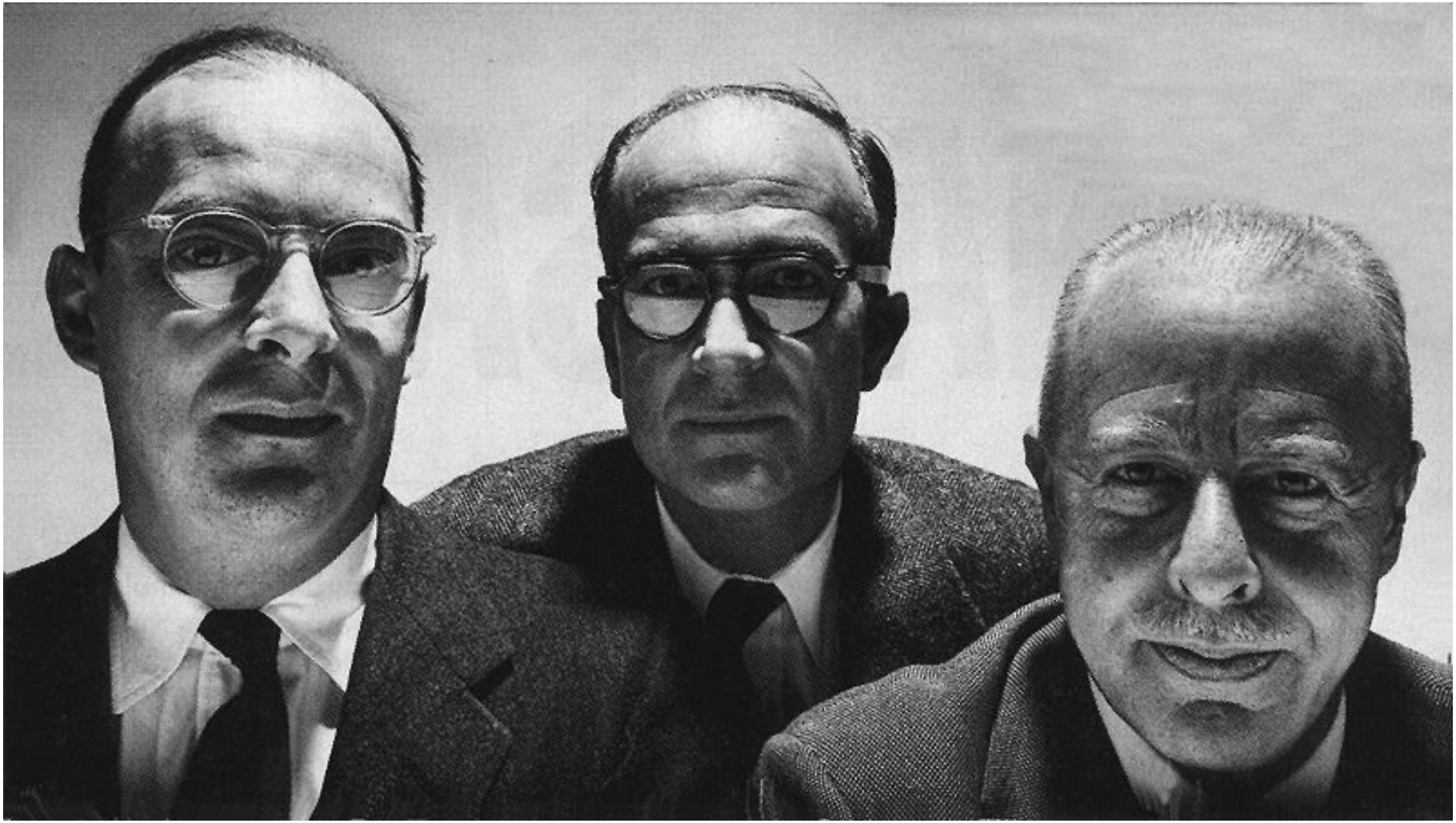
"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)



John von Neumann

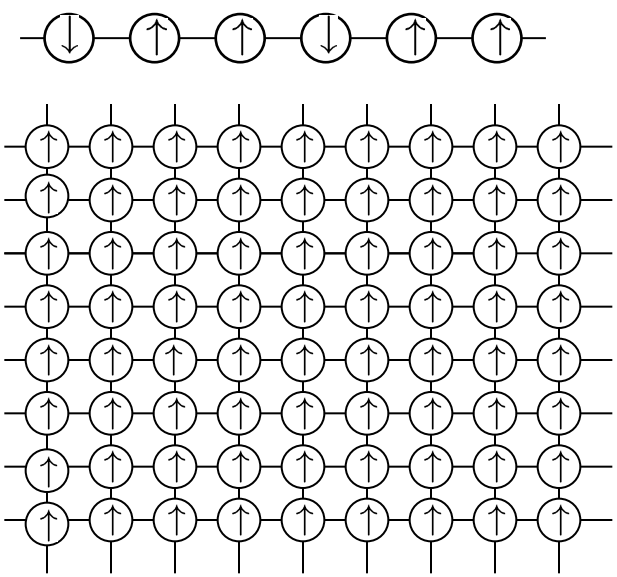
How to compute when the subsystems of your computer can fail?

A Different Path



Three Storage Testbeds

Ferromagnetic Ising models:



A. one dimensional

B. two (and >) dimensional

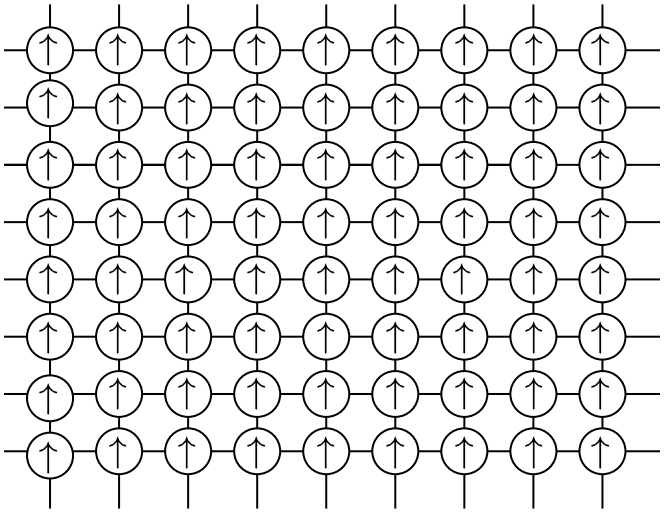
Toom's rule:

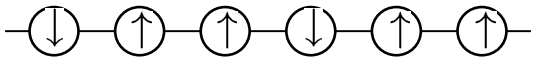
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

C. two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

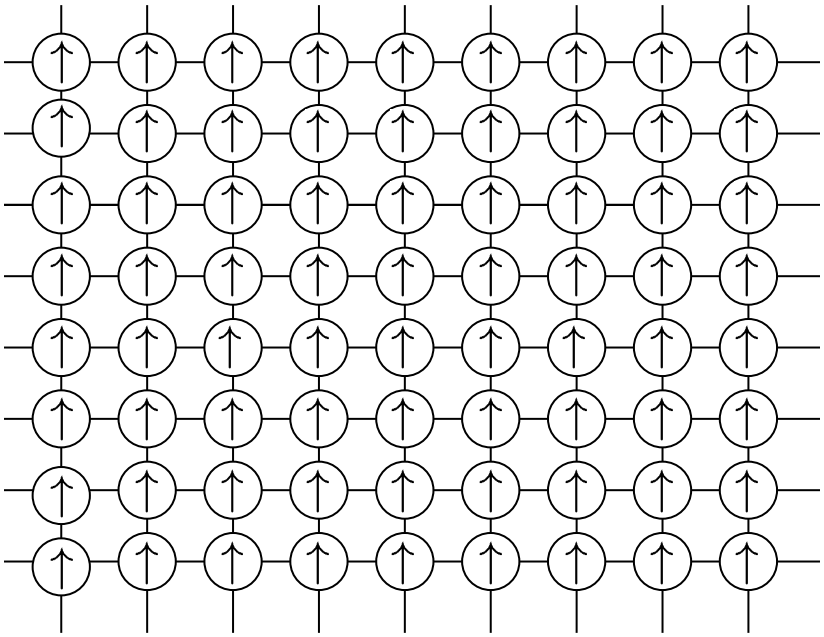
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability

$\exp(-\beta \Delta E)$



Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

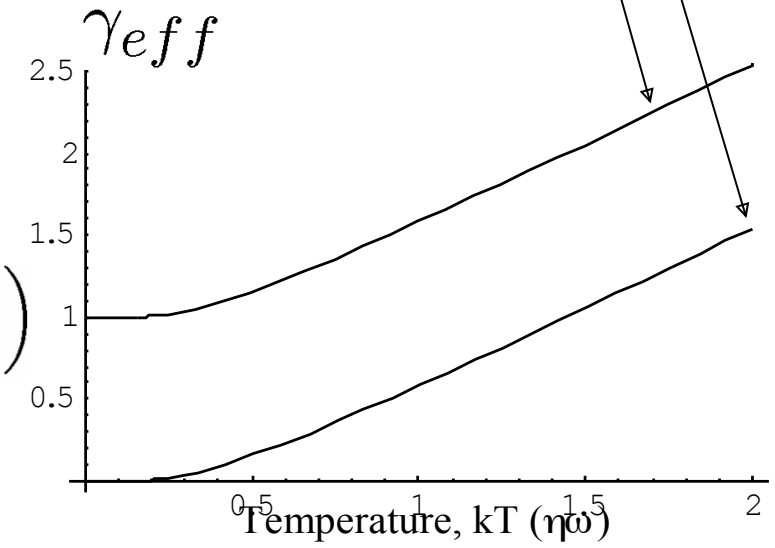
"non-dissipative"

$\hbar\omega$

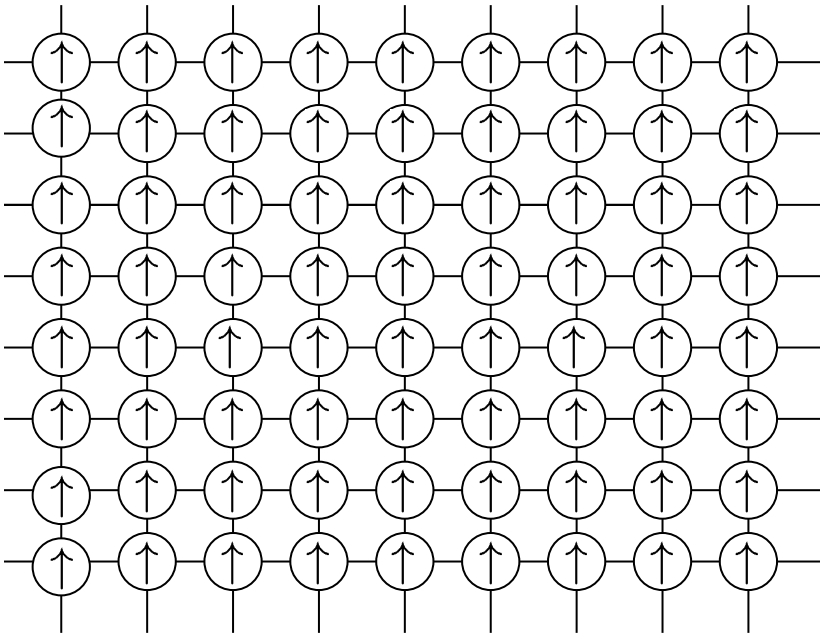
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule (alternate with noise):

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

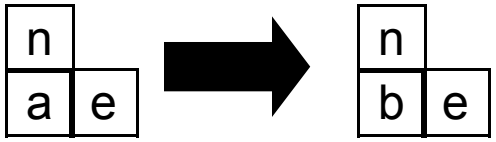
Toom's Rule



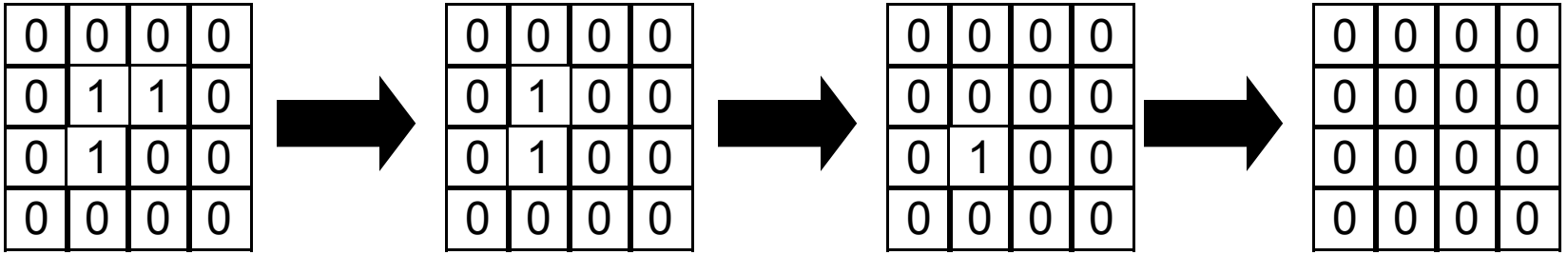
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



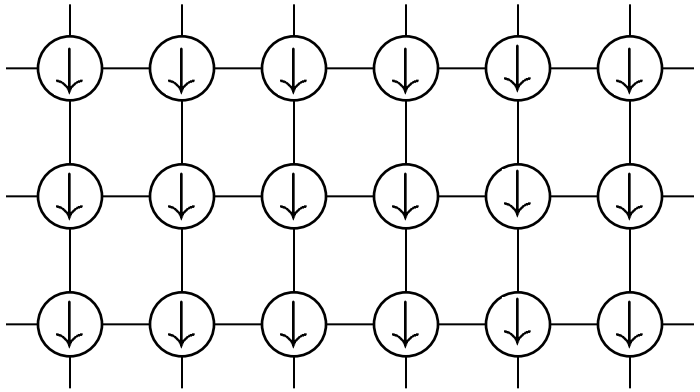
b=majority(n,e,b)



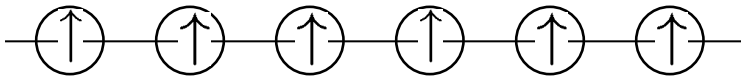
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



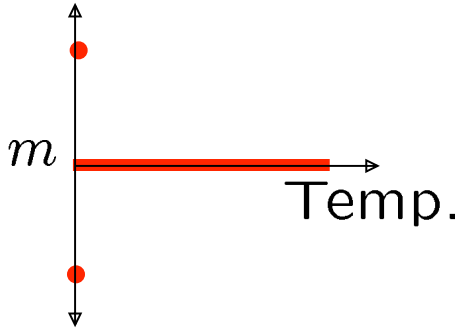
Redundancy code.

Order parameter is magnetization: $m = \frac{1}{N} \sum_i s_i$

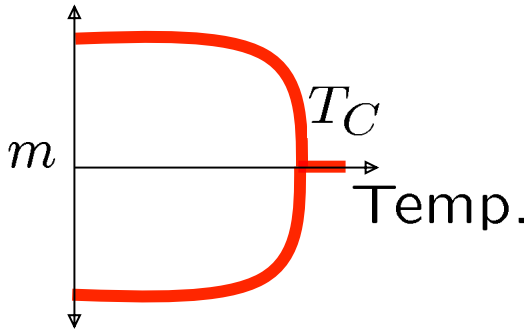
Storage

Long time behavior (not infinite time, but effectively)

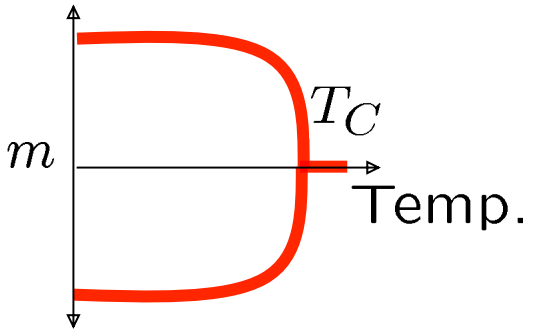
1D Ising



2D Ising



Toom's Rule



Criticism: (1) thermodynamic limit taken

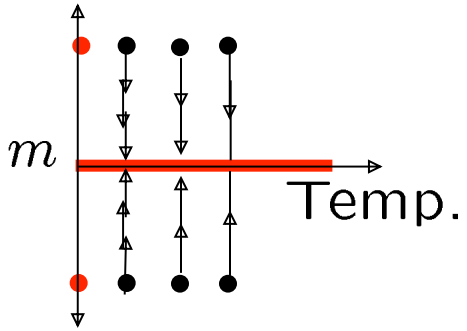
(2) what if relaxation to equilibrium takes a long time?

Storage



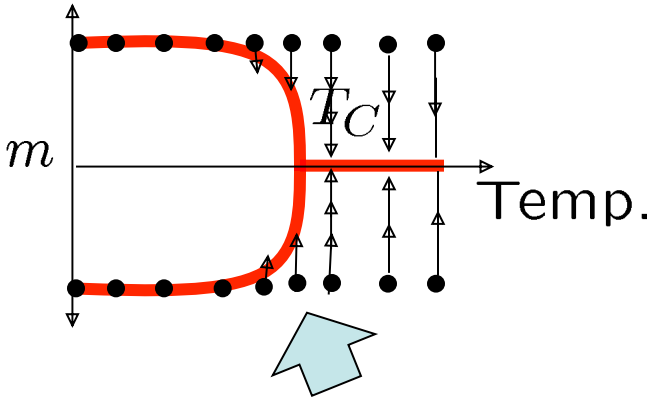
2nd attempt:
1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

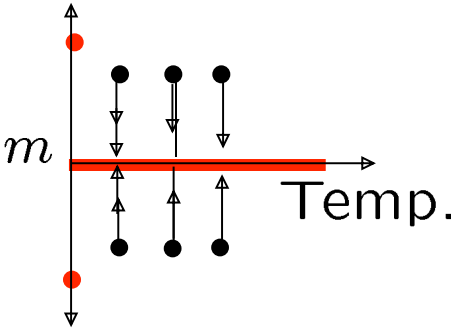
2D Ising, Toom's Rule

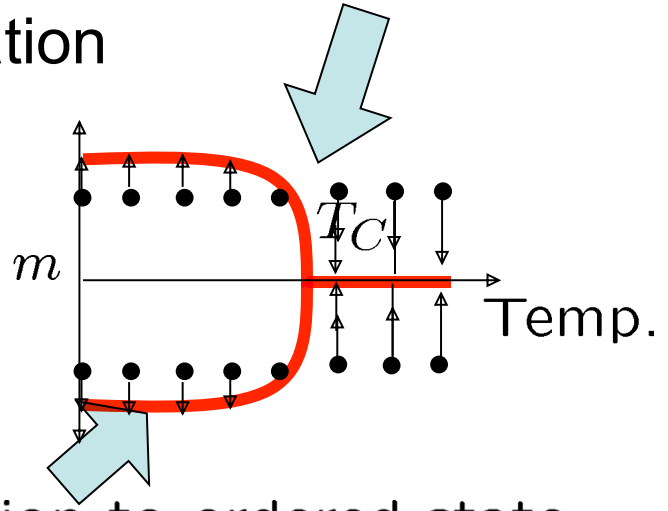


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





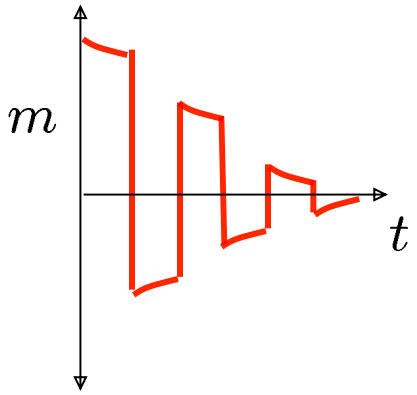
fast relaxation to ordered state

Manipulation



Flipping spins is imperfect

1D Ising

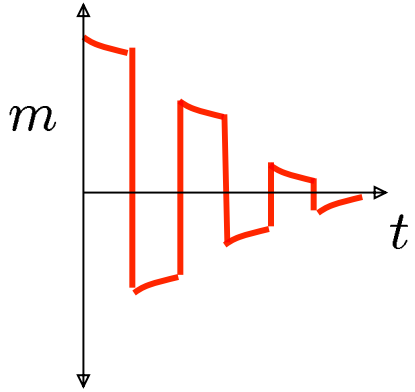


1D Ising
- temperature less than gap
- manipulation error must be small

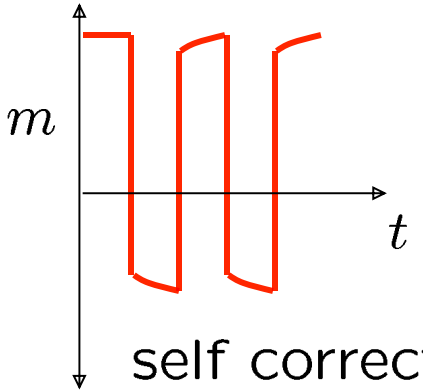
(i.e. NOT fault-tolerant)

2D Ising
- temperature less than critical temp

2D Ising
Toom's Rule



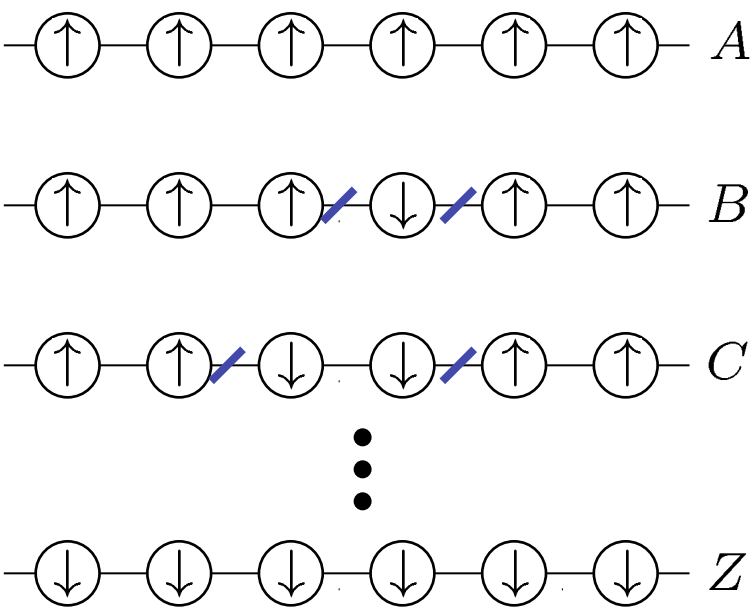
$T > T_C$

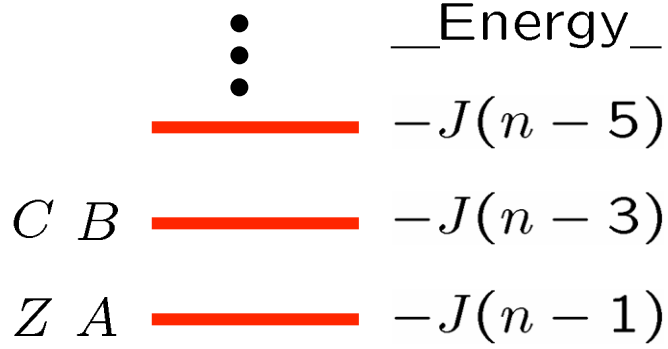


$T < T_C$

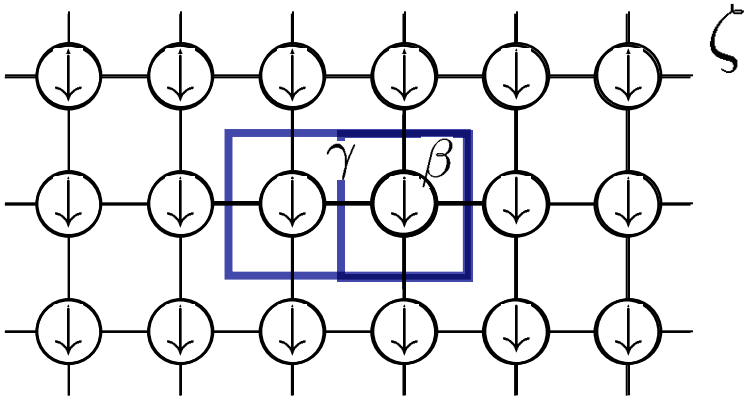
self correcting

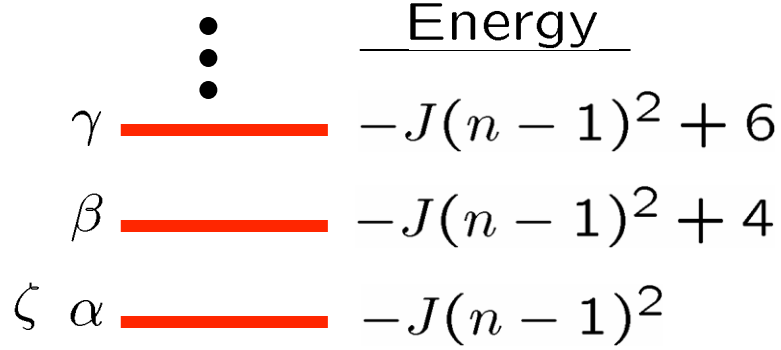
“Self Fixing”





Excitations point-like
Energy to err is constant
Disordering entropy >> Energy



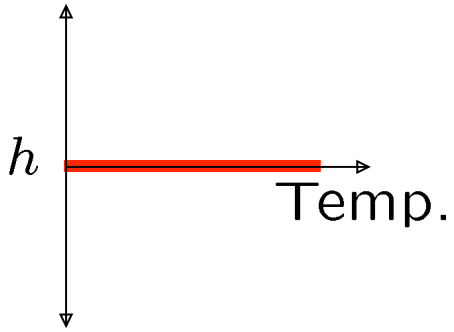


Excitations string-like
Energy to error is prop to size
Energy >> Disordering entropy

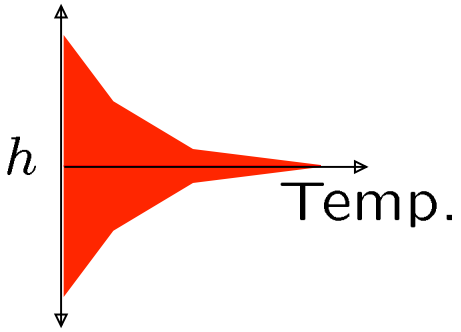
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



Toom's Rule

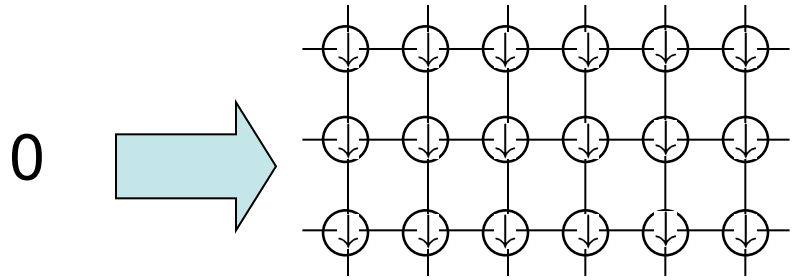


Regions with two coexisting phases

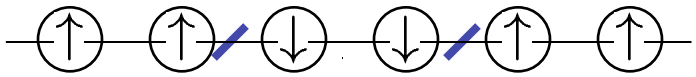
External magnetic fields disorder magnetic media

Lessons

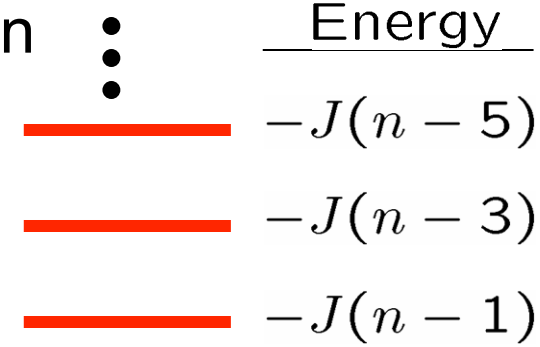
1. encode information into error correcting code



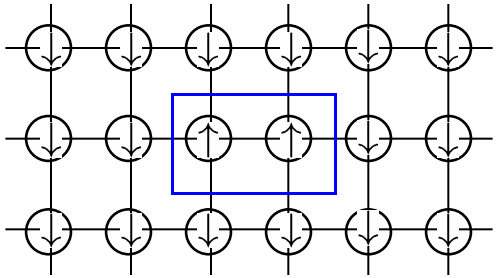
2. energy gaps can slow loss of information
   important for memory



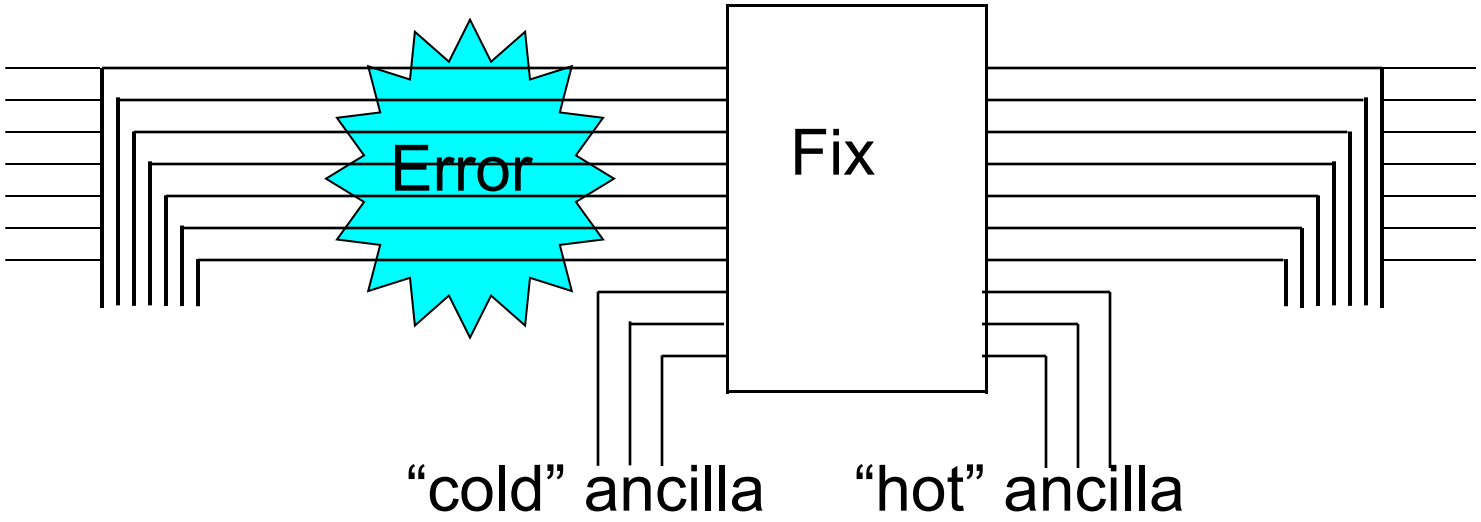
keep cold



3. excited state energetics can “self-correct”
   important for fault-tolerant operations



Quantum Error Correction

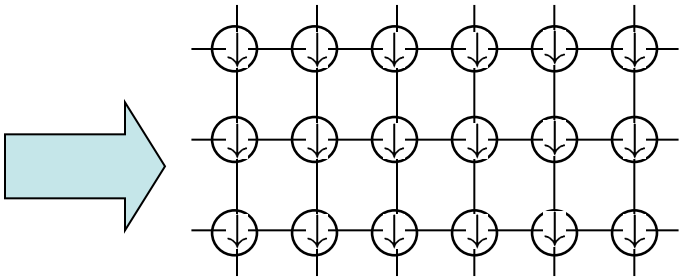


By encoding quantum information into a subspace of many quantum systems we can protect this information from noise.

Lessons Learned?

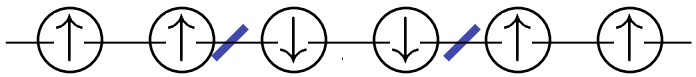
What do we do in fault-tolerant quantum computing?

1. ✓ encode information into error correcting code

0 

use information encoded into a quantum error correcting code

2. energy gaps can slow loss of information
   important for memory **NO**



keep cold ✓

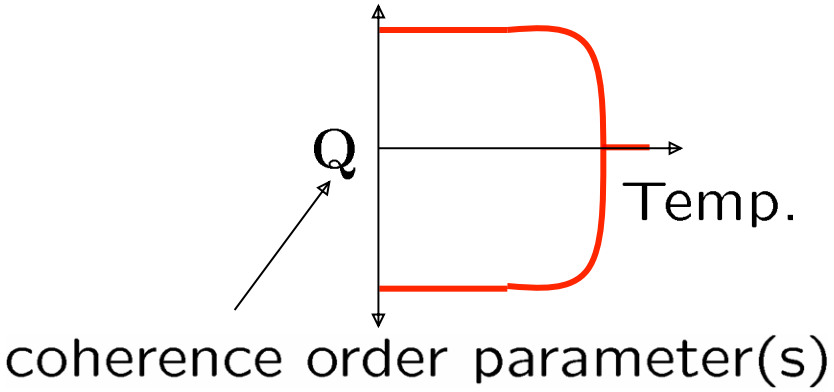
3. excited state energetics can “self-correct”
   important for fault-tolerant operations **NO**

The Quantum Hard Drive?

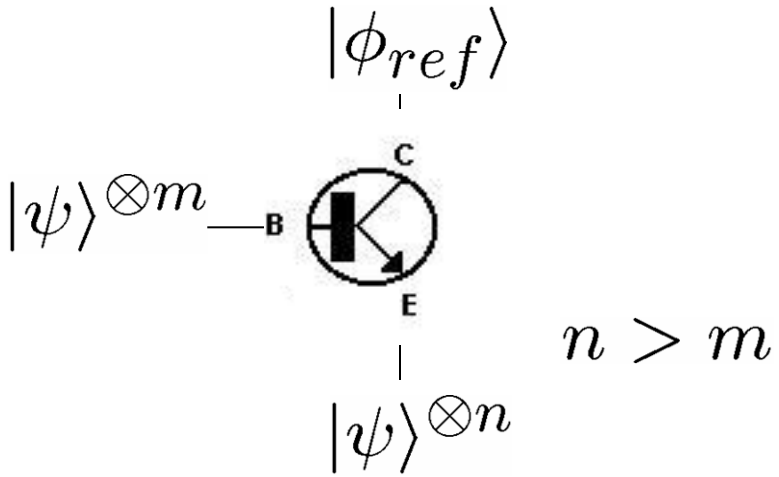


Kitaev

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



No Quantum Transistor?

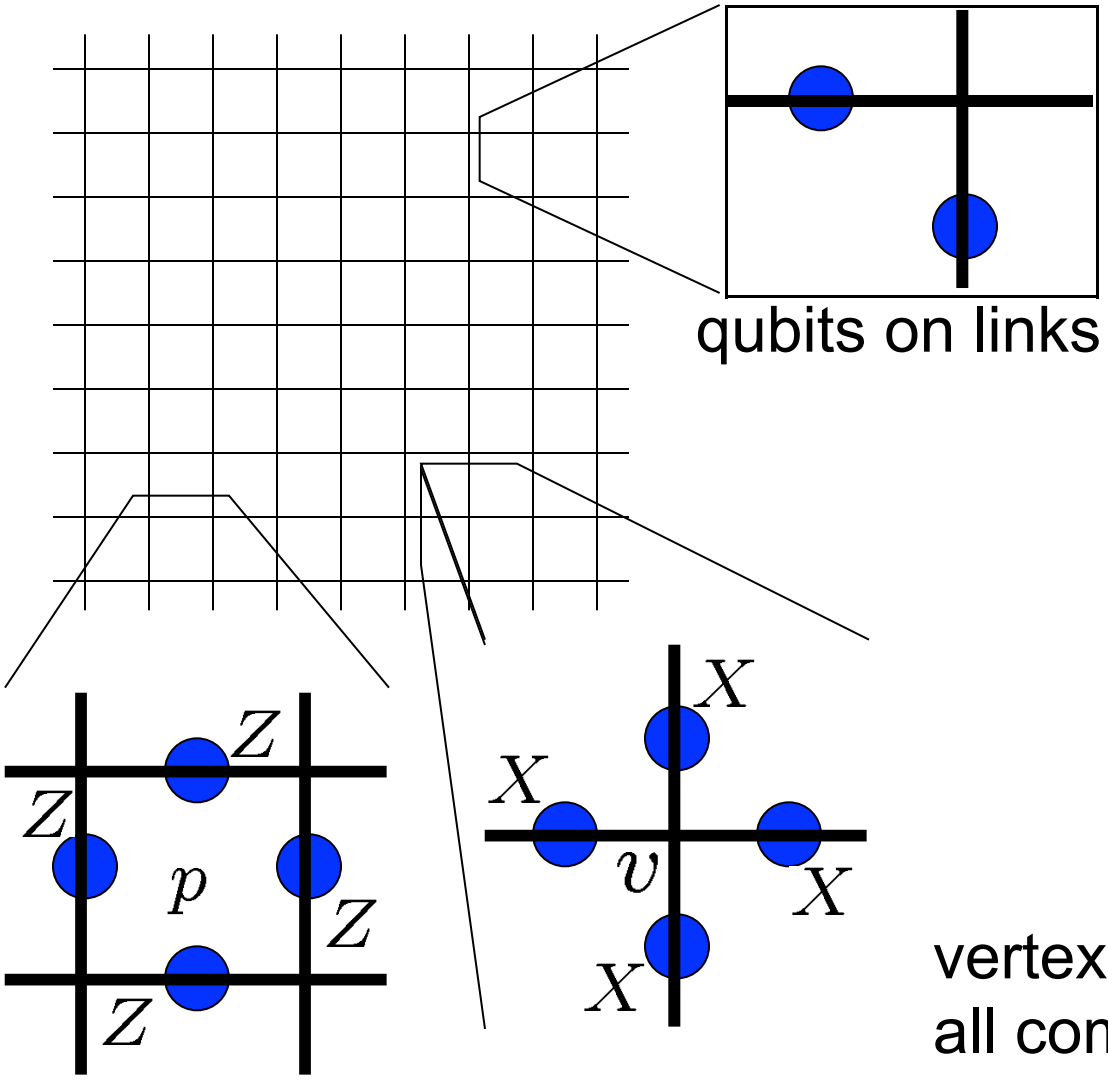




Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

Kitaev's Model

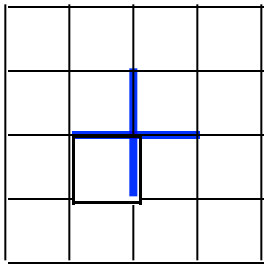


plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$



vertex and plaque operators all commute with each other

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Kitaev's Model, Continued

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

$$S_p|\psi\rangle = s_p|\psi\rangle \quad s_p \in \{-1, +1\}$$

$$S_v|\psi\rangle = s_v|\psi\rangle \quad s_v \in \{-1, +1\}$$

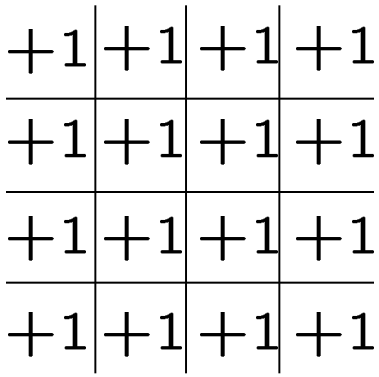
plaque operators:

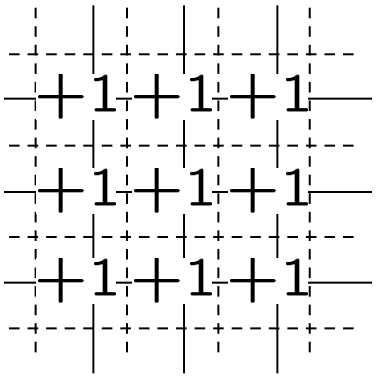
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

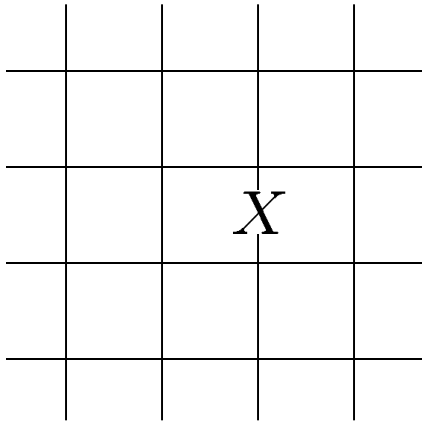
$$S_v = \prod_{e \in v} X_e$$

ground state has all $s_p = s_v = +1$





Kitaev's Model



plaque operators:

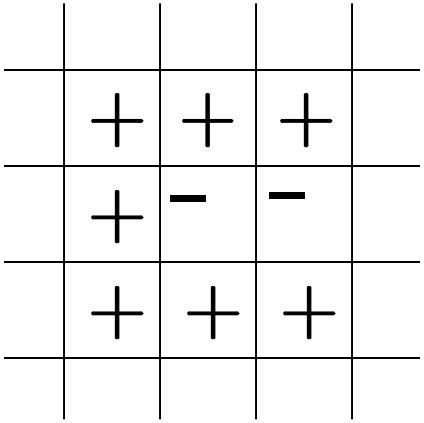
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

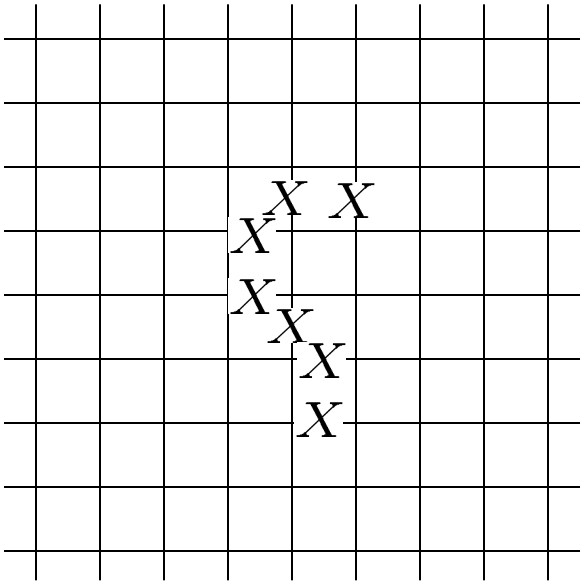
X anti-commutes with two plaque operators

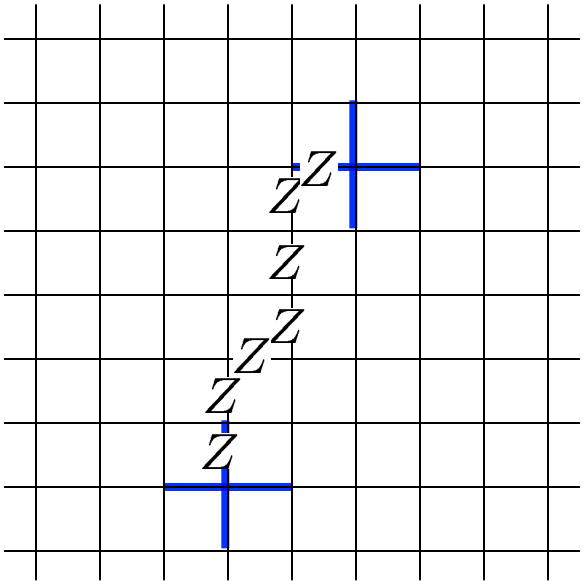
$$\{X, S_p\} = 0 \Rightarrow S_p(X|\psi_G\rangle) = -X|\psi_G\rangle$$



excitation is $4J$ above ground state

Excitations

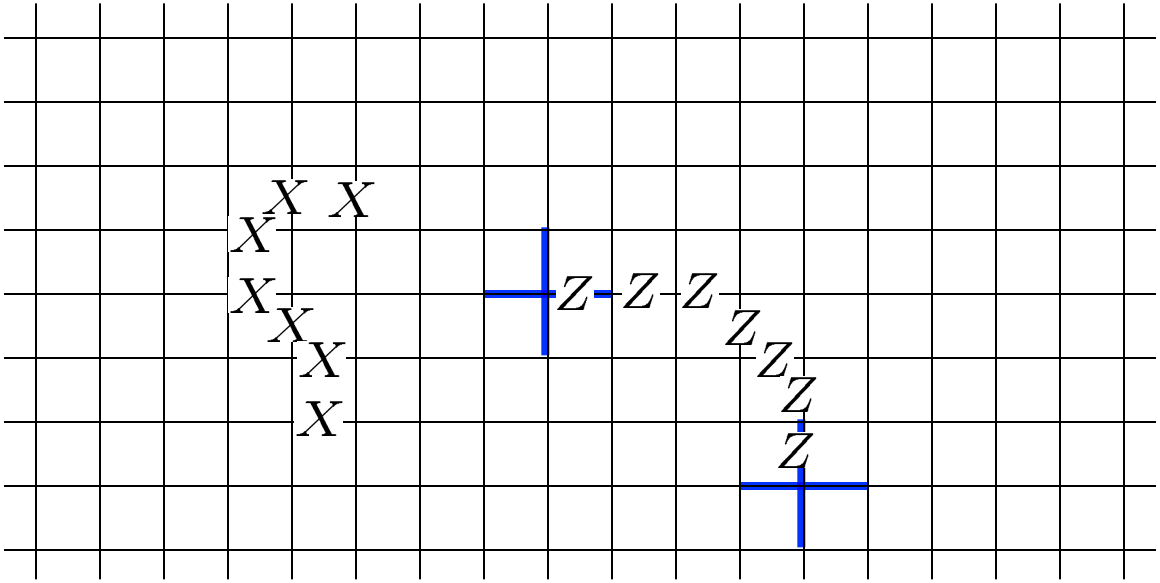




excitations particles come in pairs (particle/antiparticle)
at end of "error" chains

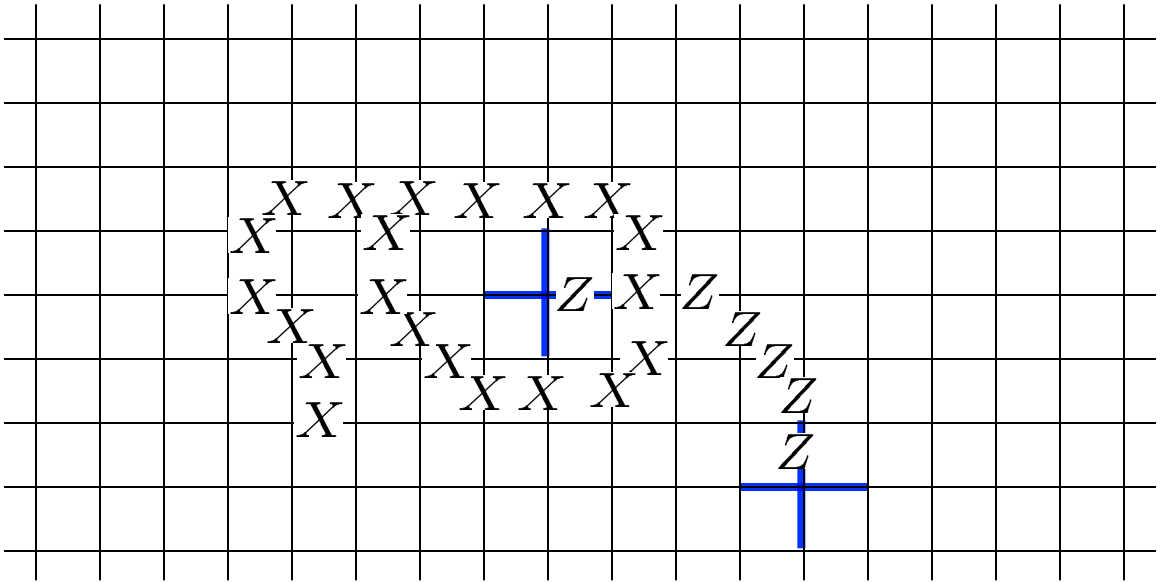
two types of particles, X-type (live on vertices of dual lattice)
Z-type (live on vertices of the lattice)

Anyons Anyone?

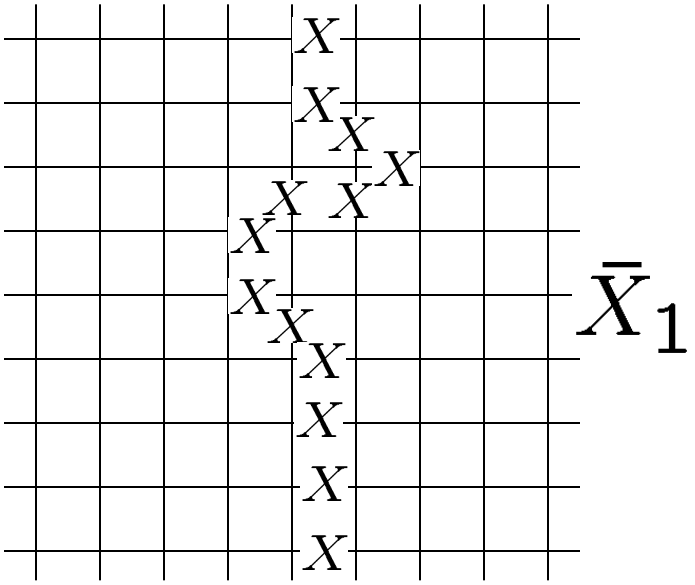


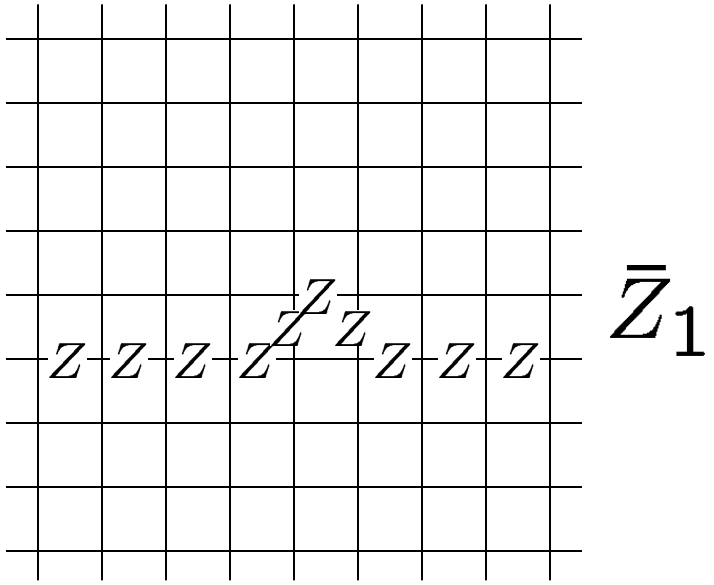
Phase:

-1



A Qubit



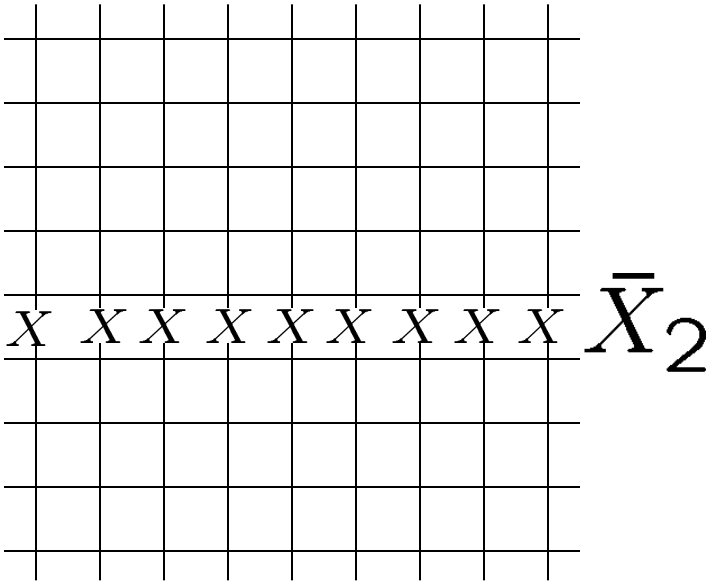


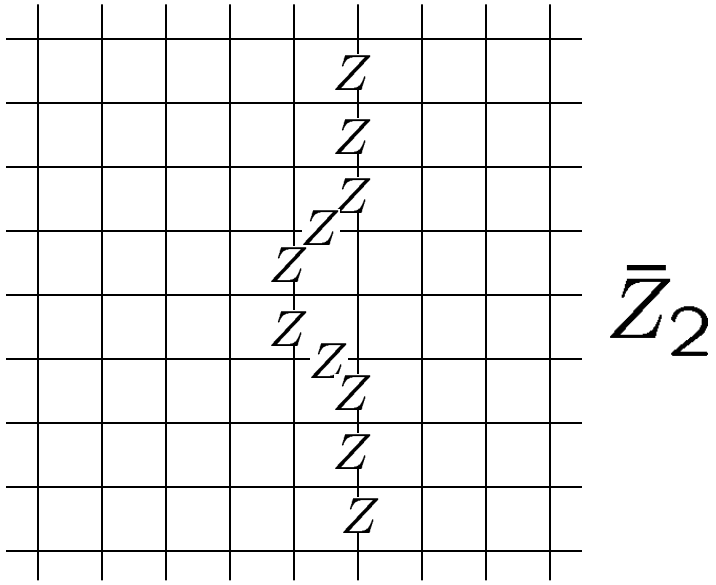
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

$$\bar{X}_1 \bar{Z}_1 \bar{X}_1 \bar{Z}_1 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Two Qubits





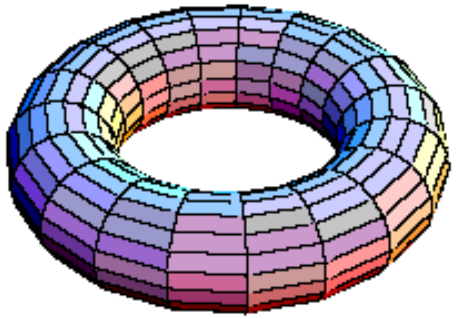
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

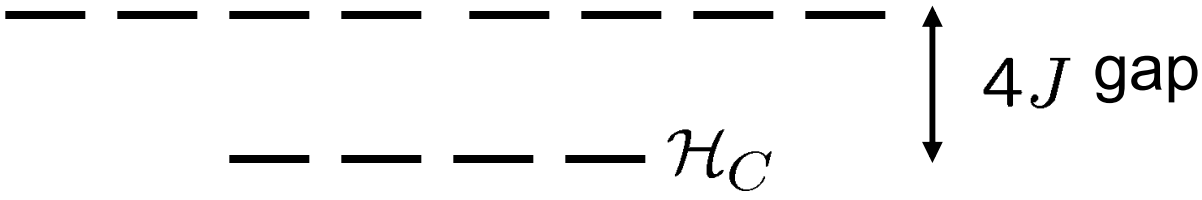
$$\bar{X}_2\bar{Z}_2\bar{X}_2\bar{Z}_2 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Topological P[illegible]

Encode two qubits
into the ground state





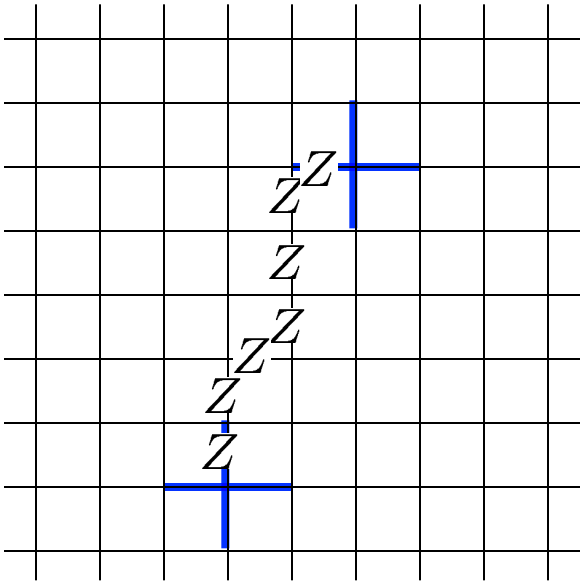
Perturbation theory:

$$V = \sum_v E_v + \sum_{vw} F_{vw}$$

But for $m \leq \left\lfloor \frac{l}{2} \right\rfloor$

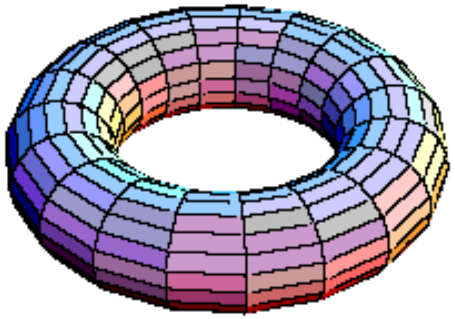
$$\langle \phi_i | V^m | \phi_j \rangle = 0$$

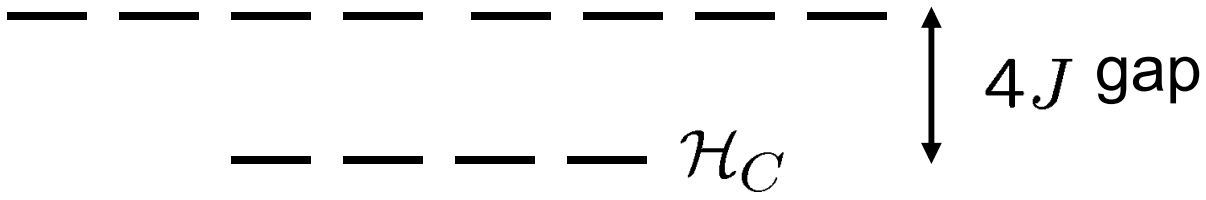
$$\langle \phi_i | V^m | \phi_i \rangle - \langle \phi_j | V^m | \phi_j \rangle = 0$$



Topological Quantum Computing

Encode two qubits
into the ground state



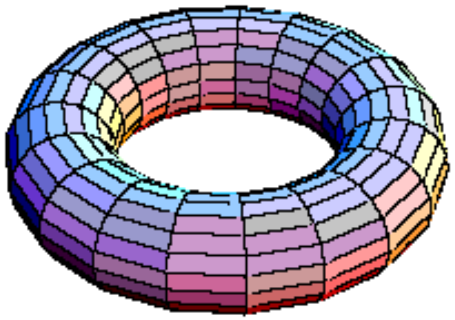


If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

Are these assumptions reasonable?

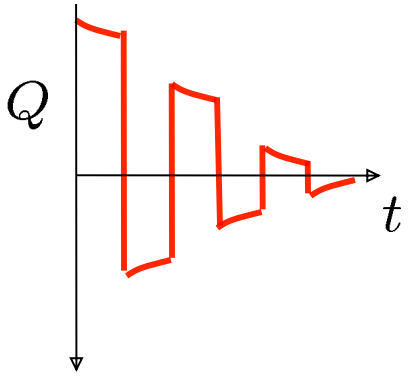
Whence Kitaev's Model?

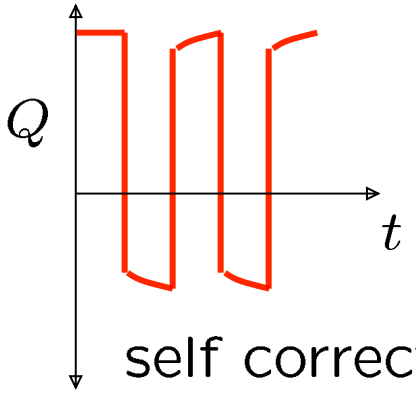


$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

obust storage of quantum information.

ut it does not provide a mechanism for
bust manipulation of quantum information.





$T < T_C$

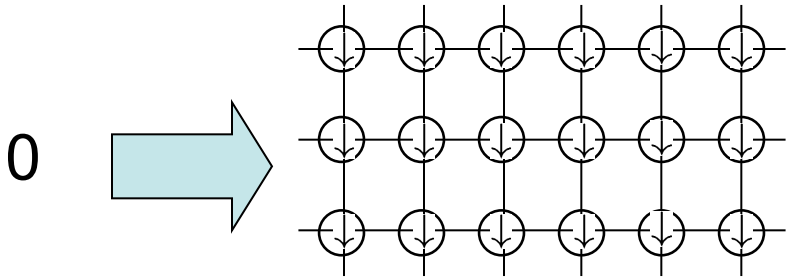
self correcting

Is it possible to construct a Hamiltonian, which allows for fault-tolerant manipulation as well?

Lessons Learned?

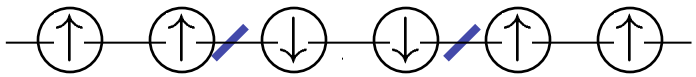
What do we do in fault-tolerant quantum computing?

1. encode information into error correcting code



use information encoded into a quantum error correcting code

2. energy gaps can slow loss of information
   important for memory

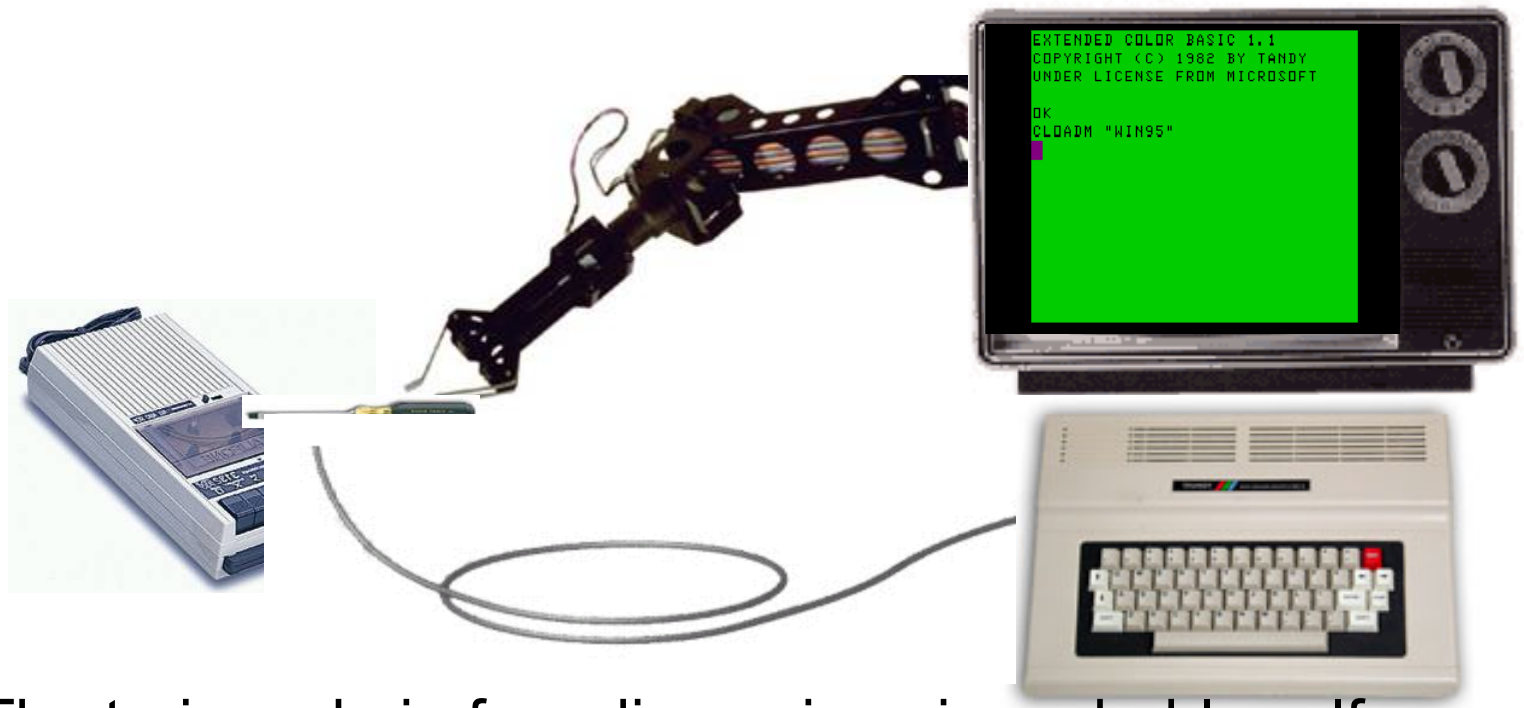


keep cold

3. excited state energetics can “self-correct”
   important for fault-tolerant operations

NO

Self Correcting Quantum Computers

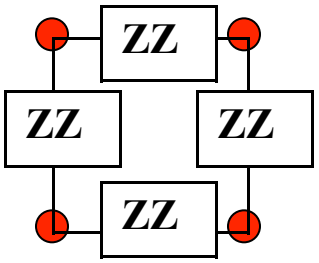


The toric code in four dimensions is probably self-correcting

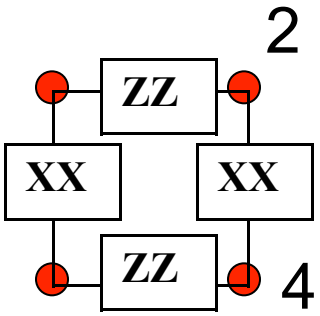
Can we self-correct in realistic systems?

D. Bacon “Operator Quantum Error Correcting Subsystems for Self-Correcting Quantum Memories”
Phys. Rev A, 73, 012340 (2006)

A Small Quantum Try



classical



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$
$$S_2 = X_1X_2X_3X_4$$

symmetry

$$[H, S_1] = [H, S_2] = 0$$

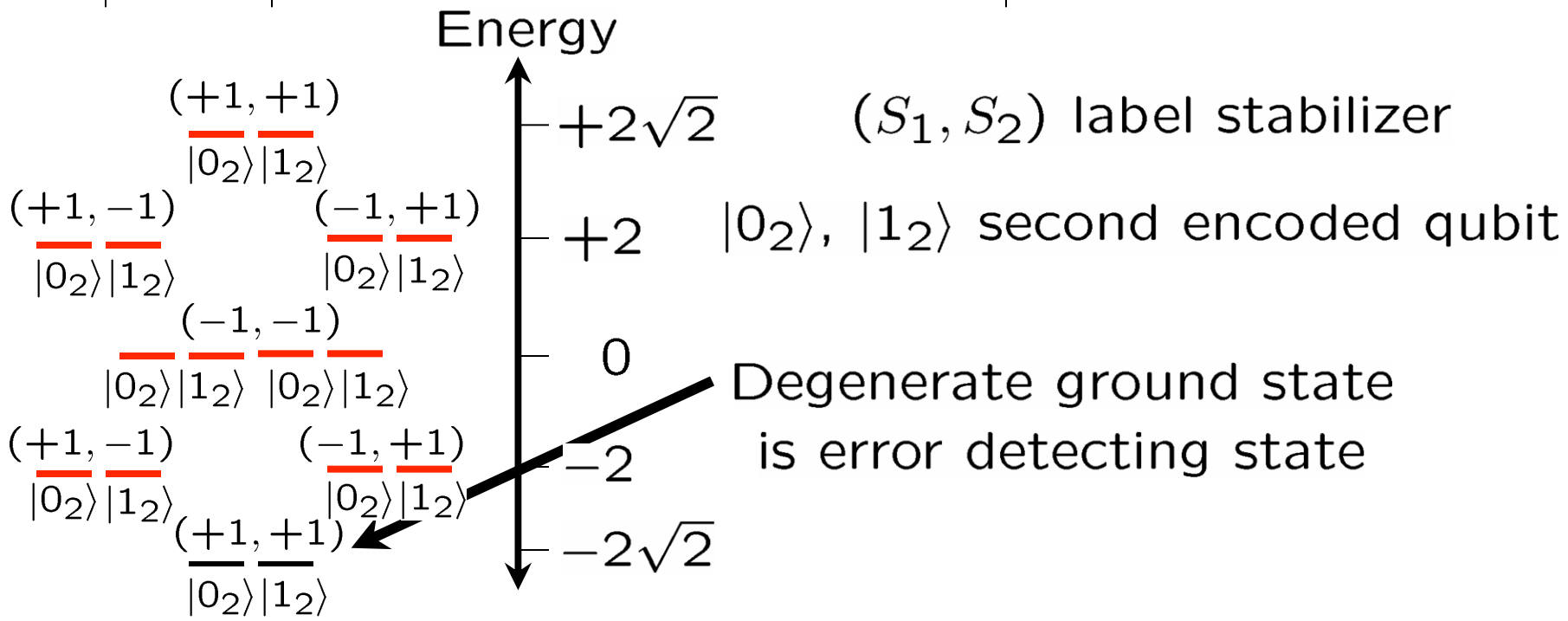
$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$
$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

simultaneously diagonalize with symmetries

H does not depend on second qubit

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0





Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

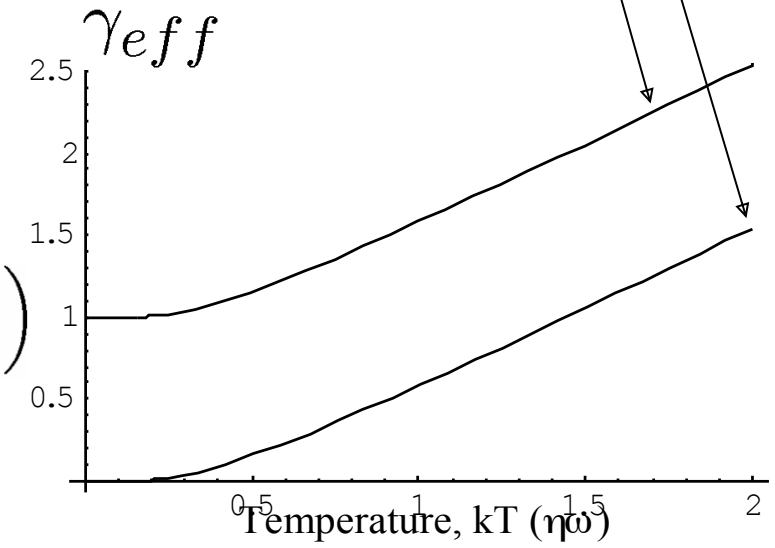
"non-dissipative"

$\hbar\omega$

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Supercoherence

Bacon, Brown, and Whaley, Phys. Rev. Lett. **87**, 247902 (2001)

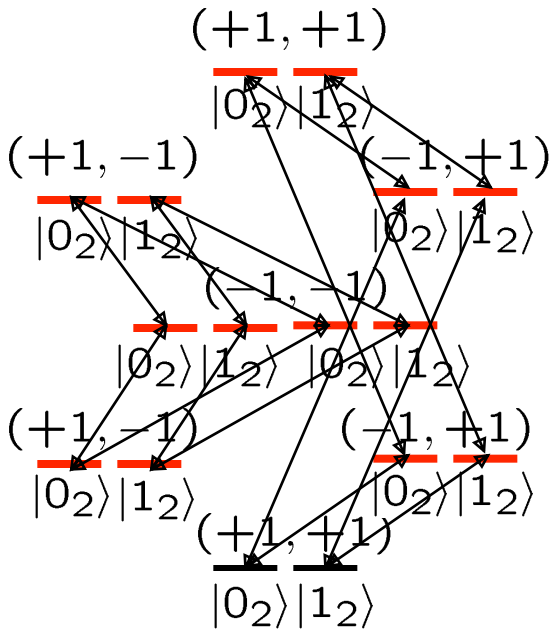
$$H = -(Z_1 Z_2 + Z_3 Z_4 + X_1 X_3 + X_2 X_4)$$

$S_i P = -P S_i$ for some i $\quad P$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**



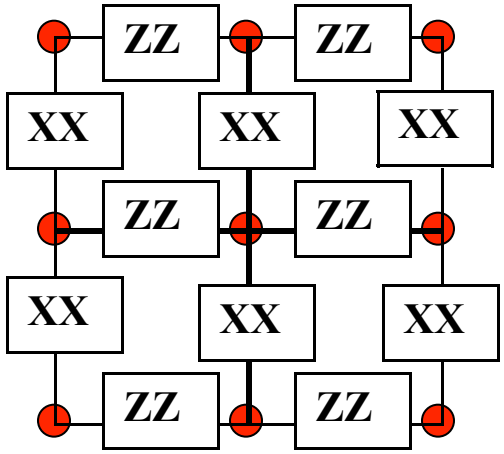
$$\underbrace{(+1, +1)}_{|0_2\rangle} \qquad \underbrace{\qquad}_{|1_2\rangle}$$

✗ **FORBIDDEN**

To first order all single qubit decoherence becomes heating

Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

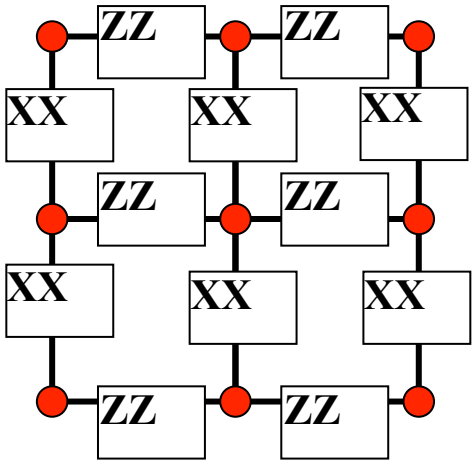


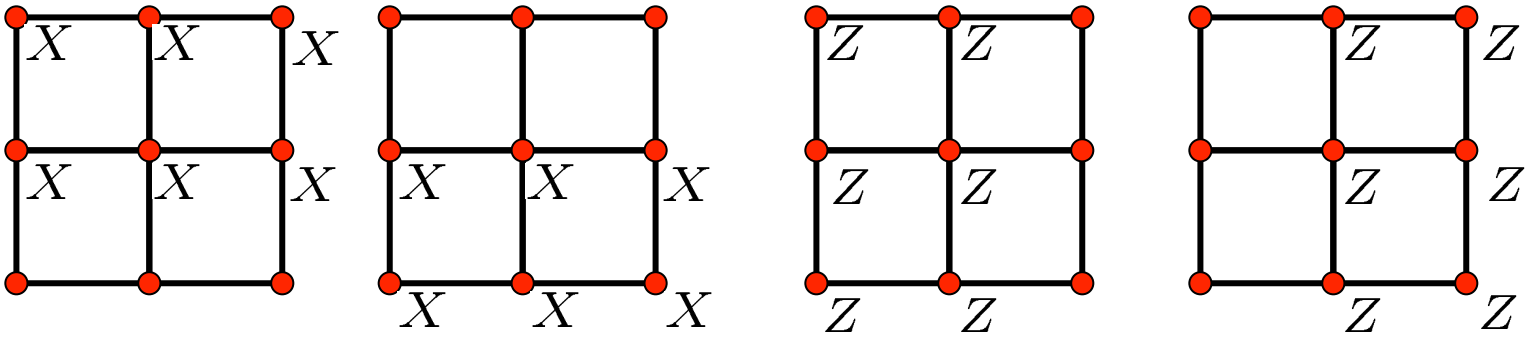
Ground state is a new kind of quantum error correcting code!

quantum error correcting subsystem

Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$





Abelian symmetries

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two qubits

$$\mathcal{H}_2 \otimes \mathcal{H}_2$$

<u>encode a qubit in a subspace</u>

$$|0_L\rangle = |1\rangle \otimes |1\rangle$$

$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$\alpha|0_L\rangle + \beta|1_L\rangle$$

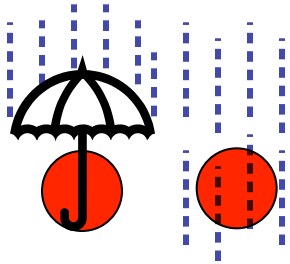
<u>encode a qubit in a subsystem</u>

$$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$$

generic subsystem:

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

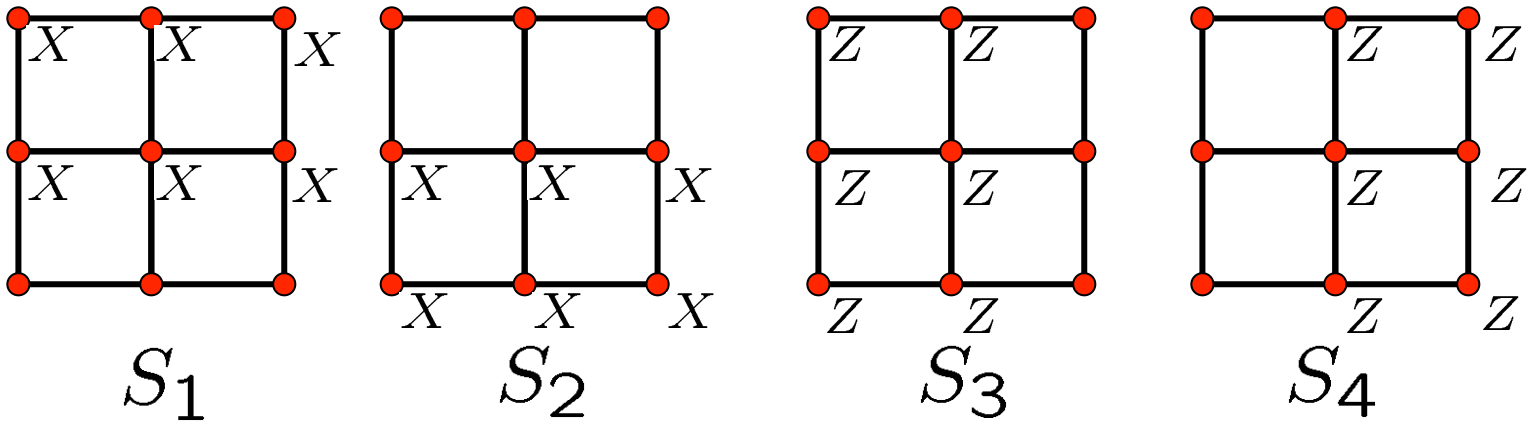
$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0$ — $\mathcal{E}$ — $\mathcal{R}$ — $(|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$

error recovery

encoded quantum information

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Quantum Subsystem Code



. Measure these operators

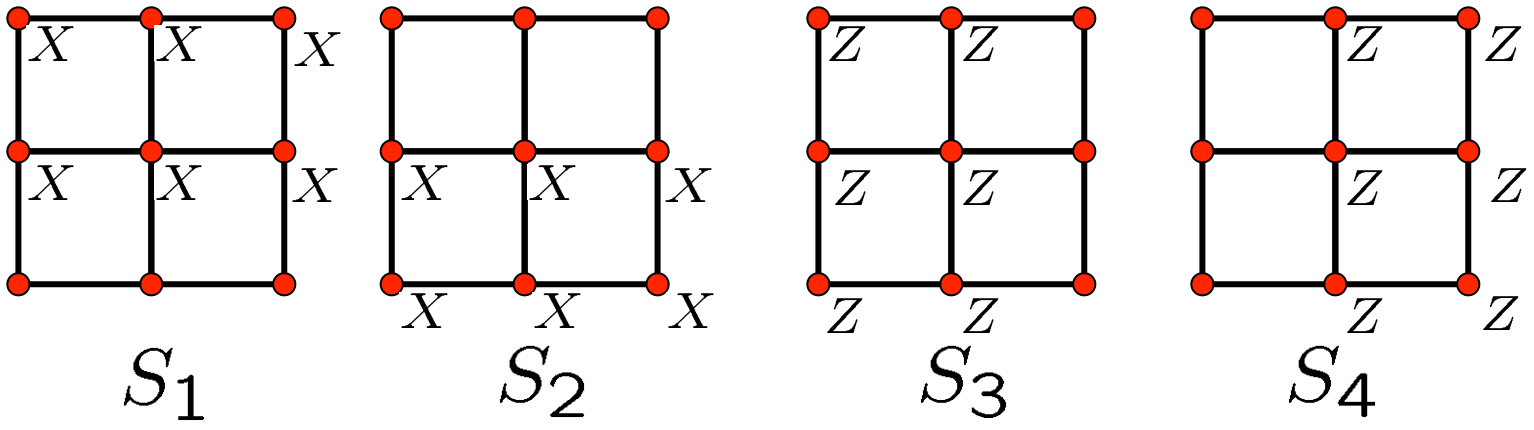
. Diagnose parity of bit and phase flips in column and rows

3. Apply recovery routine for parity of errors

Routine fixes error UPTO an operation generated by bonds of Hamiltonian

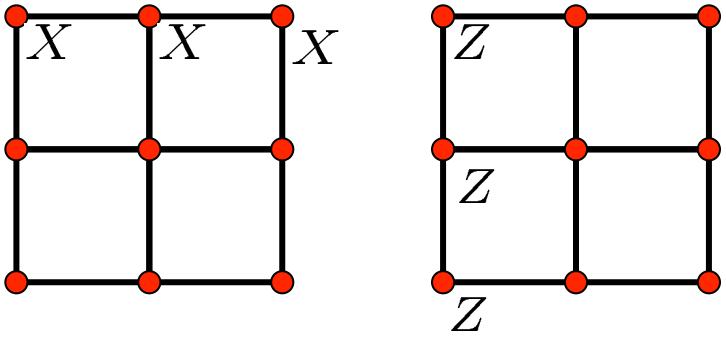
$$X_{i,j}X_{i+1,j}, Z_{i,j}Z_{i,j+1}$$

Quantum Subsystem Code



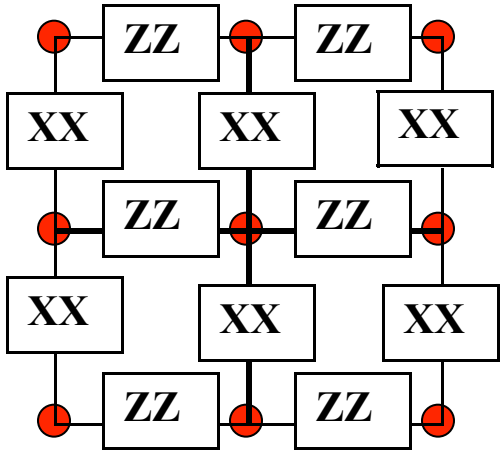
$$X_{i,j}X_{i+1,j},\, Z_{i,j}Z_{i,j+1}$$

extra symmetry which implies states are doubly degenerate:



Quantum Compass Model

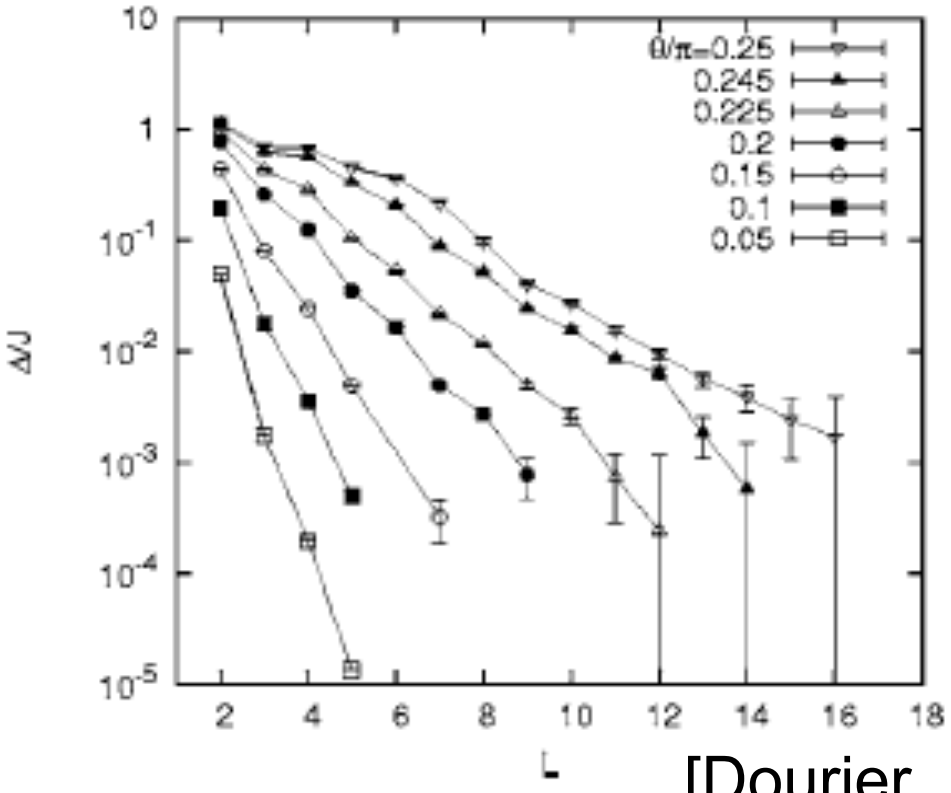
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is a new kind of quantum error correcting code!

quantum error correcting subsystem

Problems:

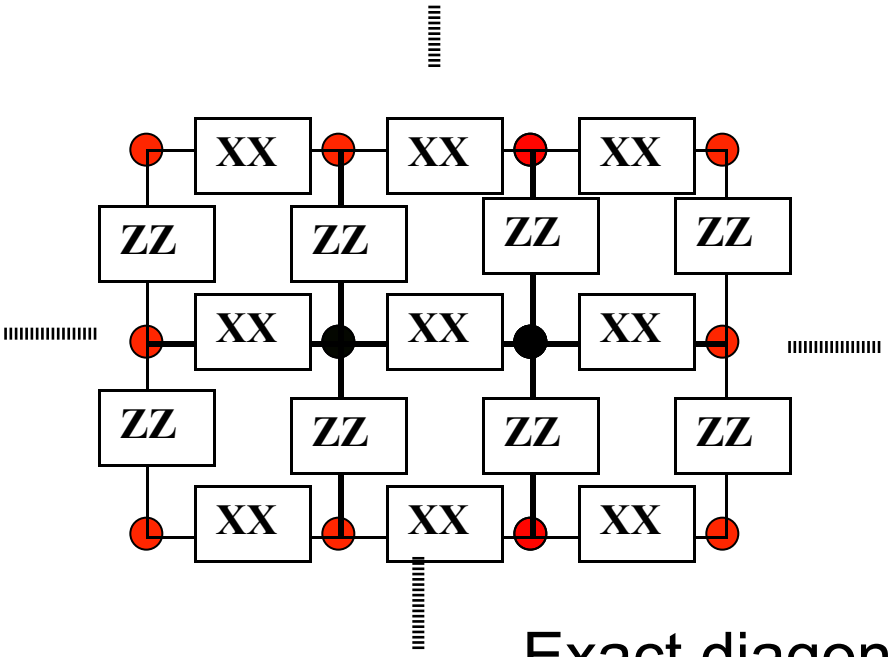


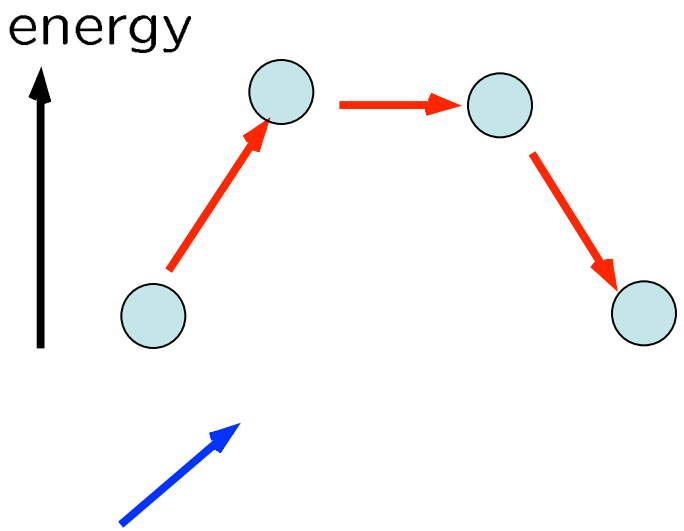
[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

Problems, Continued



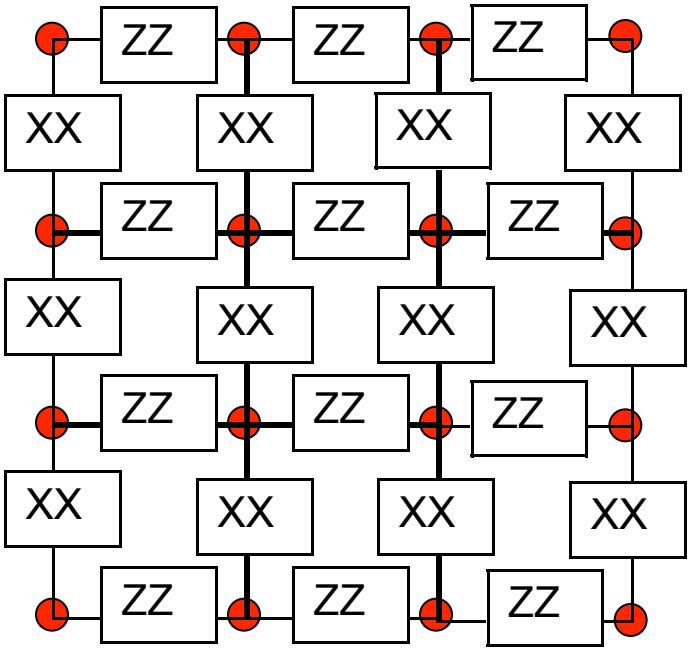


Exact diagonalization 3x3, 4x4, 4x5

Even at finite size, not “self-correcting”

Toward Self-Correction

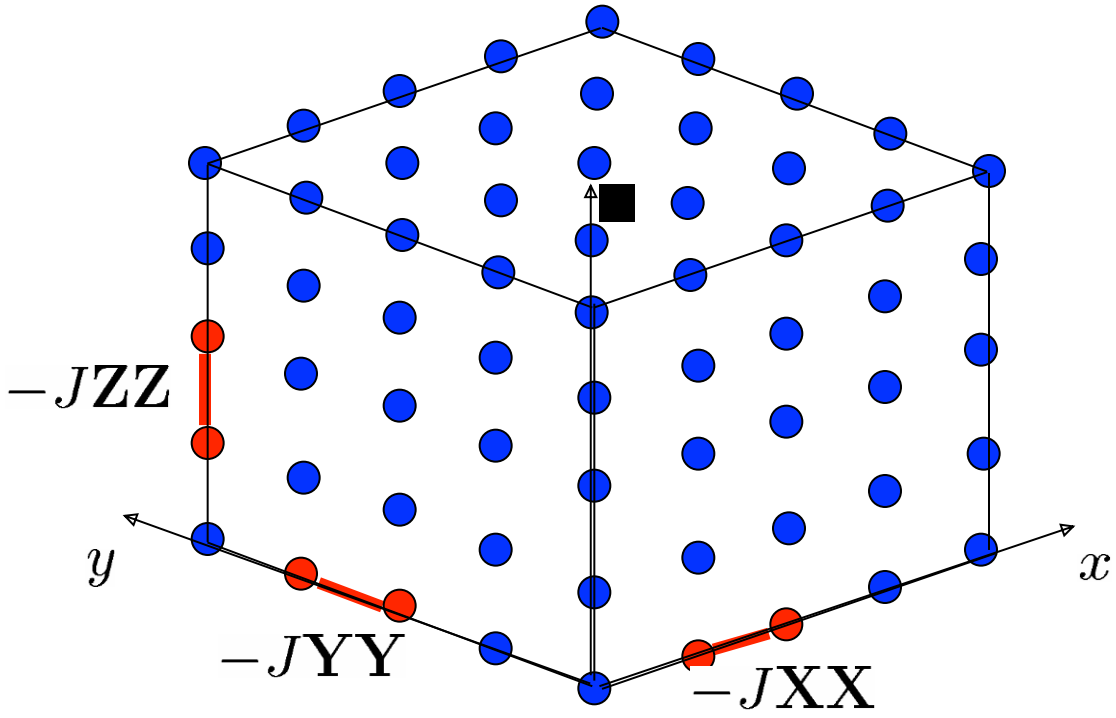
n by *n* square lattice



$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Exact diagonalization of 4x4 model, shows that with constant couplings, this model is self-correcting!

Toward Self-Correction



$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

The Future

Can techniques used in creating two qubit interactions be used to engineer such interactions
(Ideas of Deng, Porras, and Cirac?)

Possible to engineer such systems in planar ion trap system?

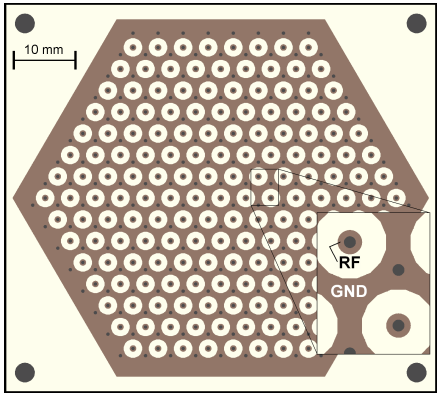
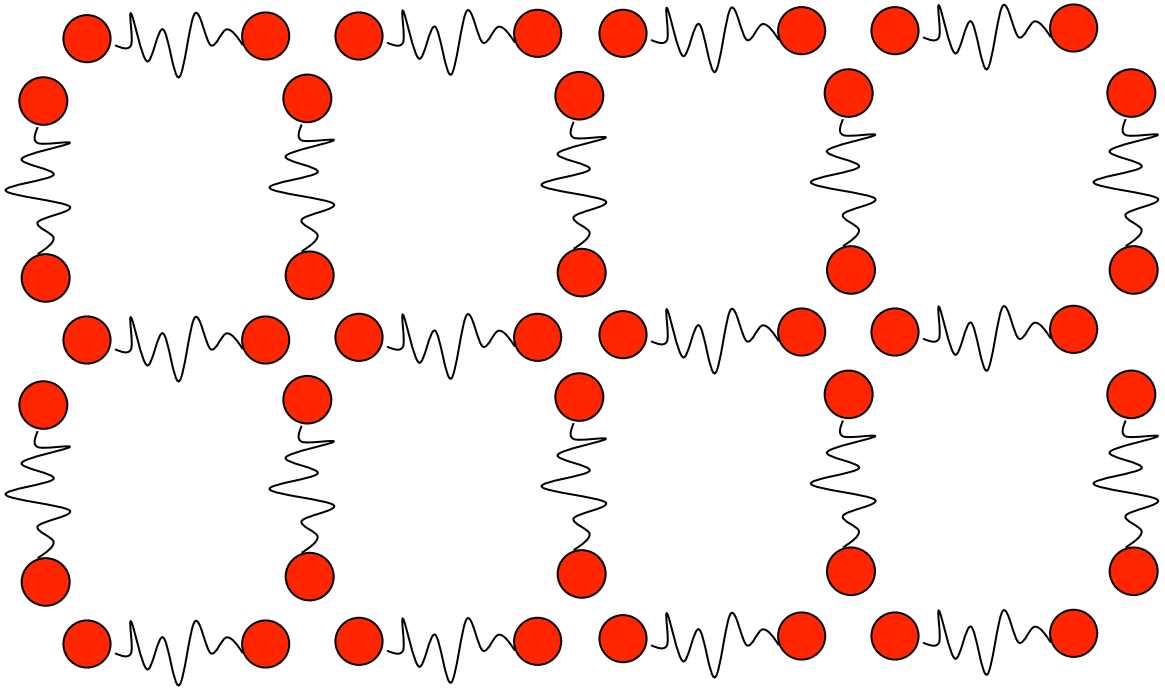


Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.

Picture courtesy Ken Brown
(Isaac Chuang's Lab, MIT)

The Future

systems with “little” entanglement are efficiently simulatable



projected entangled pair states (Verstraete and Cirac)

simulation of self-correcting models

A Different Path



Quantum error correction can be performed by cooling an interacting quantum system.

Potential for robust substrates for quantum information.